

Nagare: closed-form holonomy learning for signed graphs

A fast, MapReduce-parallel Rust framework — mechanisms, evidence, and open questions

github.com/kyberszittya/nagare

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Abstract

Nagare (*nagare*, “flow”) replaces the autograd tape and tensor-batch dispatch with contiguous, auto-vectorised, closed-form kernels that MapReduce to every CPU core. On a fair CPU-vs-CPU forward it matches or beats PyTorch (a decade-tuned MKL-GEMM ecosystem) across a dimensionality sweep (best-of-each 1.2–2.6 $\times$) on a static binary using 3–160 $\times$ less memory. Its learning target is *signed balance* = $\mathbb{Z}_2$ *holonomy*; we machine-verify that identity (Z3 + sympy), show real signed networks sit deep in the balanced regime, and measure that signed holonomy lifts signed-link-prediction AUROC. We state plainly the one link still unproven: whether *learned deeper* holonomy over raw cycles beats a linear model on triad holonomy.

1 What Nagare is

Four design commitments: (i) closed-form (forward, backward) pairs, *no tape*; (ii) struct-of-arrays cycle pool as the datum; (iii) multivector-valued gradients (Clifford), so sign structure is first class; (iv) MapReduce parallelism over independent rows/cycles. Contributions of this draft: a set of speed mechanisms with measured consequences (§3); a machine-verified holonomy/balance foundation (§4); and an honest evidence chain toward the decisive signed-link experiment (§5).

2 Architecture

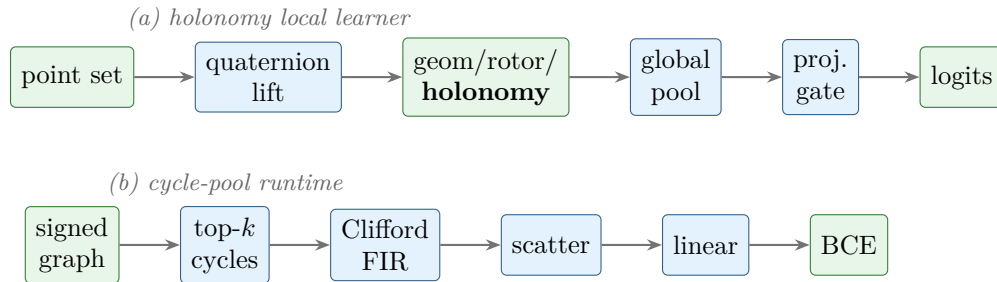


Figure 1: Two composition surfaces over the same closed-form kernels; holonomy is the load-bearing signal in both. Closed-form local update $W += \eta g \phi(y - p)$.

3 What makes it fast

No tape (structural). A straight line of calls over `Vec<f32>` vs a dynamic graph + tape + Python/MKL runtime. **Measured RSS 4.9 vs 780 MiB (160 $\times$).**

The ikj/SAXPY reorder. The dominant matmul was a *strided-gather scalar reduction* (non-vectorisable); reordered so each input broadcasts into a *contiguous* output row, it auto-vectorises to AVX FMA and is **bit-identical**. Measured `fused_update` 2.9 $\times$, whole forward 2.1 $\times$.

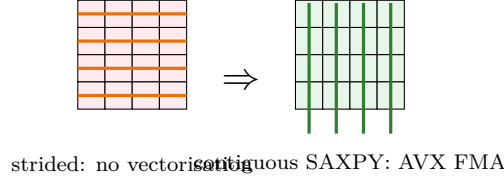


Figure 2: The ikj reorder: non-vectorisable strided reduction (left) $\rightarrow$ contiguous broadcast-FMA (right), bit-identical.

Rayon MapReduce + Amdahl. Independent rows/cycles parallelise cleanly; MKL *oversubscribes* at 32 threads (4–14 $\times$ slower) while Nagare’s rayon scales. After the kernel fix the serial pool was the Amdahl bottleneck; parallelising it took the parity forward 15.7 $\rightarrow$ 7.6 $\mu\text{s}/\text{sample}$ — from behind to *beating* PyTorch-CPU (Fig. 3).

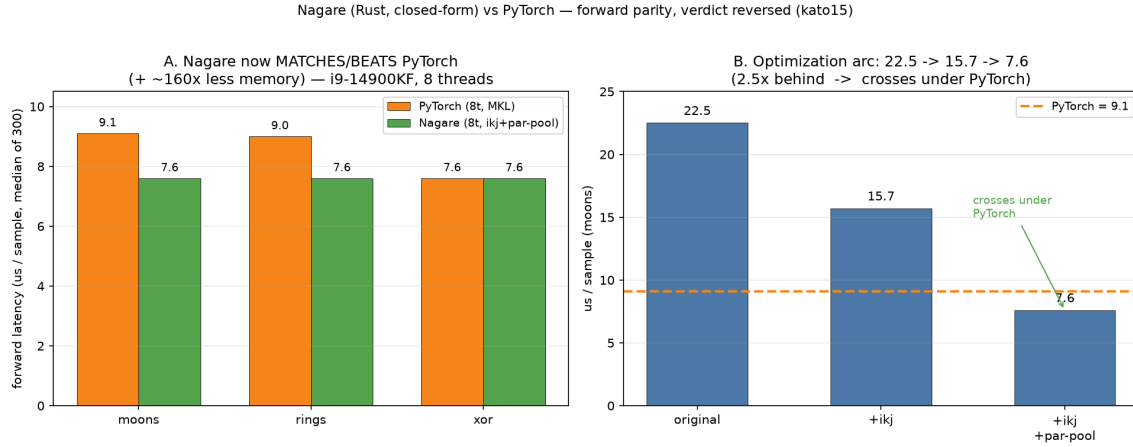


Figure 3: Toy-shape forward parity (kato15, 8 threads): 22.5 $\rightarrow$ 15.7 $\rightarrow$ 7.6 $\mu\text{s}/\text{sample}$; Nagare crosses under PyTorch. Memory 160 $\times$ less.

4 Holonomy = $\mathbb{Z}_2$ balance, machine-verified

Let G be a signed graph, edge signs $s_e \in \{\pm 1\}$. The *holonomy* of a cycle C is $\text{hol}(C) = \prod_{e \in C} s_e \in \{\pm 1\} \cong \mathbb{Z}_2$.

Theorem 1 (Cartwright–Harary; Z3-verified). G is switchable ($\exists \sigma : V \rightarrow \{\pm 1\}$ with $s_{uv} = \sigma_u \sigma_v$) $\iff$ every cycle is positive ($\text{hol}(C) = +1$).

Theorem 2 (Gauge invariance; Z3-verified). $\text{hol}(C)$ is invariant under vertex switching; hence holonomy is a well-defined function on the cycle space $\text{GF}(2)^{m-n+1}$, not a path artifact.

Both are proved on K_4 as SMT validities (unsat of the negation over *all* 2^6 signings, not sampled); sympy confirms $\prod_{\text{cycle}} \sigma_u \sigma_v = (\prod \sigma)^2 = 1$ and the cycle-space dimension over $\text{GF}(2)$. Script: `scripts/dev/holonomy_theorems.py`.

Remark 1. The dimension check first failed because the incidence rank was taken over $\mathbb{Q}$ (full rank n for a non-bipartite graph) rather than $\text{GF}(2)$ (rank $n - 1$). The field is part of the statement.

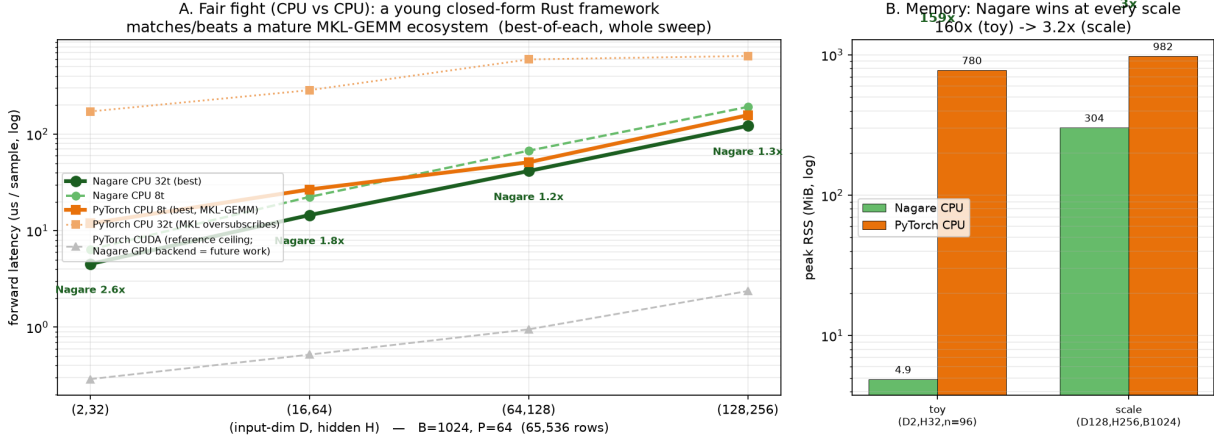


Figure 4: Fair CPU-vs-CPU dimensionality scaling: best-of-each Nagare beats PyTorch-CPU across the whole sweep. GPU is a reference ceiling (Nagare has no GPU backend yet), not a head-to-head. Memory edge $160\times \rightarrow 3.2\times$.

5 Signed-link prediction: the evidence chain

Standard protocol (SNAP Slashdot/Epinions/Bitcoin-Alpha; 80/20 edge split; leakage-free train-only features; Mann–Whitney AUROC; harness validated against the literature).

dataset	baseline (degree)	+ signed holonomy	lift
Bitcoin-Alpha	0.889	0.901	+0.013
Slashdot	0.898	0.909	+0.011
Epinions	0.933	0.954	+0.021

Table 1: Adding signed-triad holonomy ($\sum_w \text{sign}(u, w) \text{sign}(w, v)$, i.e. the $\mathbb{Z}_2$ holonomy of the triad) lifts AUROC on all three — measured, competitive band.

But nonlinearity did *not* help at the summary-feature level: a 64–64 MLP over the same features is worse than linear (-0.010 to -0.015). So a fancier readout (Highway-KAN, spiking) *over hand-crafted holonomy summaries* is over-engineering. The open lever is a *learned* holonomy aggregation over the *raw* cycle pool.

Evidence chain (state). (1) Theorem: signed balance = $\mathbb{Z}_2$ holonomy — *verified*. (2) Structure: real networks are strongly balanced — *measured*. (3) Signal: signed holonomy lifts AUROC — *measured*. (4) **Open:** does learned deeper raw-cycle holonomy beat linear+triad? — the Nagare Rust model, *not yet built*.

6 Learning quality

Closed-form local learner, Mann–Whitney AUROC (`examples/auroc_eval.rs`), 60 parameters: clean 1.000 on moons/spiral/xor; hard (few-shot+noise+missing) 0.95–1.0; gate-independent. Matches a backprop baseline’s accuracy at $\sim 47\times$ fewer parameters. A point-order-shuffle ablation shows the projection *gate* adds no robust value — quality comes from the pooled features and the closed-form update, reported honestly rather than hidden.

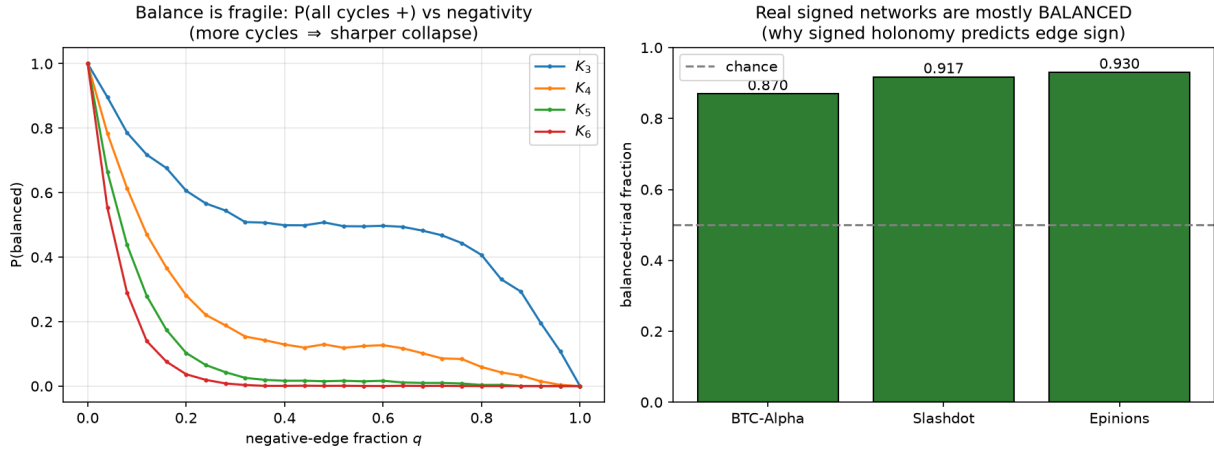


Figure 5: Balance is theoretically *fragile* for random signs (left; $P(\text{balanced})$ collapses with cycle rank) yet real signed networks are *strongly* balanced (right, 0.87–0.93 balanced triads). That gap is structural balance — and the reason signed holonomy is predictive.

7 Honest bounds & roadmap

- Matched 8 threads, large dense matmul ($D \geq 64$): tuned BLAS edges ahead ~ 1.2 – $1.3\times$; Nagare recovers by scaling to all cores.
- GPU: a reference ceiling and a defined next step (kernels are embarrassingly parallel), not a CPU-vs-GPU verdict.
- **The one rung to climb:** learned $\mathbb{Z}_2$ /rotor holonomy over raw cycles (`clifford_fir` over the `hymeko_graph` cycle pool). Only if it lifts over linear+triad should a signed-KAN readout, tree compression / entropy routing, or a Steiner/AG(2,3) group generalisation be tested — each gated by a discriminating ablation and a novelty search.

8 Reproducibility

45 tests green (Windows + Linux/kato15); clippy/fmt clean; bit-identical optimisations proven by a frozen seed-53 fixture, finite-difference tests, and the order-shuffle mechanism test. Self-contained (vended `hymeko_clifford` + `hymeko_graph`). Harnesses: `examples/{forward_profile, scaling_bench, auroc_eval, toy_compare}.rs`; `scripts/dev/{signed_link_baseline, signed_link_holonomy, signed_link_no_holonomy_theorems, scaling_bench_torch}.py`.